Use-Case Flow Specification

Public Bus Live Tracking & Crowding Estimator — Problem Statement #24

Use Case ID	UC-01
Use Case Name	View Live Bus Status
Primary Actor	Commuter
Related Requirements	FR-002 (primary), FR-003 (via included UC-04), FR-001/NFR-002 (via included UC-03)
Preconditions	• The Commuter has access to the transit platform (mobile app or web client). • The requested route, stop, or bus exists in the system's route/schedule data. • Current or recently received GPS telemetry (FR-001) and ticketing sensor data (FR-003) are available for at least one bus serving the selection.
Postconditions	• Success: The Commuter's screen displays the current location, ETA, and crowding level for each relevant bus, consistent with the most recent telemetry ingested by the system. • The system has recorded the request for monitoring/analytics purposes without altering any bus, route, or sensor data.
Main Success Scenario	1. Commuter selects a route, stop, or specific bus in the client application. 2. System retrieves the current route/stop information and identifies the buses relevant to the selection. 3. System executes UC-03 (Calculate Bus ETA) — included — to obtain the current ETA for each relevant bus, using the latest GPS data ingested under UC-02. 4. System executes UC-04 (Estimate Passenger Crowding) — included — to obtain the current crowding level for each relevant bus, using the latest ticketing sensor data. 5. System assembles the bus location, ETA, and crowding level into a single live-status view. 6. System displays the live bus status to the Commuter. 7. Commuter views the live bus status. 8. Use case ends successfully.
Alternate Flow	**3a. GPS data unavailable or stale for a bus** • 3a1. During step 3, the system detects that no GPS update has been received for a bus within the last 15 seconds (three missed 5-second ingestion cycles per FR-001). • 3a2. System marks that bus's location and ETA as “unavailable” rather than displaying a stale value as current. • 3a3. System displays the affected bus with a clear “Live data unavailable” status instead of a numeric ETA. • 3a4. Commuter may continue viewing any other available buses for the selected route/stop; the use case resumes at step 5 for those buses.

Notes: UC-01 includes UC-03 (Calculate Bus ETA) and UC-04 (Estimate Passenger Crowding) per the UML use-case diagram — both are required, not optional, sub-behaviours of viewing live bus status, which is why they are modelled with ««include»» rather than ««extend»».